

NEWS & VIEWS

INFORMATION THEORY

A signal take on speech

Michael S. Lewicki

Approaches that abandon traditional speech categories offer promise for developing statistical descriptions that encapsulate how speech conveys information. Grandparents would be among the beneficiaries.

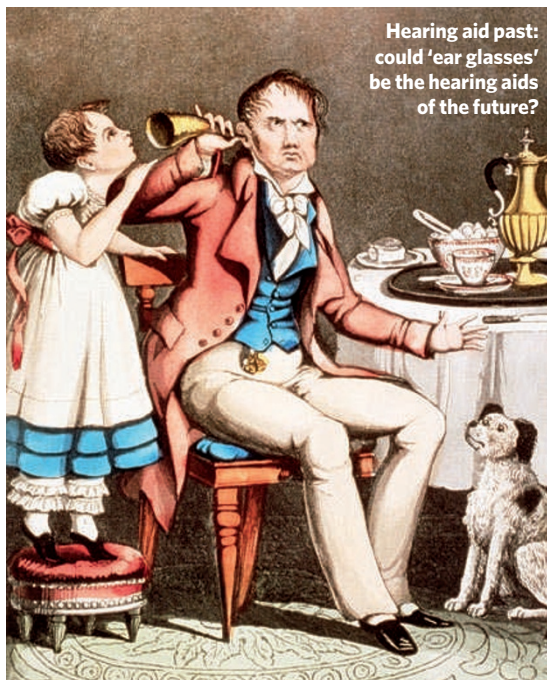
Our own ease of understanding speech belies its underlying complexity. At an abstract level, it is easy enough to describe speech as a sequence of words or of phonemes, but it's notoriously difficult to analyse at the level of the acoustic signal. We are also very good at understanding speech in noisy environments, a problem that remains vexing for both speech-recognition systems and users of hearing aids. As reported in their paper in *Proceedings of the National Academy of Sciences*¹, Stilp and Kluender have applied the methods of information theory to offer insights into the intelligibility of speech in the presence of noise.

It goes without saying that speech carries information, but what are the basic elements of the code? Which elements carry the most information? Or, alternatively, can speech be broken down into features that are equally informative, and, if so, what would those be?

A phoneme is usually taken to be the smallest unit of speech sound that can distinguish the meaning of one utterance from another ('bill' and 'dill', for instance). However, a phoneme is not a single sound, but a whole distribution of sounds, the variation of which does not change the meaning. This variation makes it difficult to quantify the information content and to understand the factors that contribute to intelligibility.

The analogous problem with text is far easier. For example, consonants are more informative than vowels simply because there are more of them and each one occurs less often. You c•n•nd•rst•nd t•xt q•st• w•ll w•th•t v•w•ls, because speech is redundant and the consonants carry enough information to determine the words. In spoken speech, there are 15 vowels and diphthongs (in American English) and still more consonants (25 or 26), but vowels are typically longer than consonants and have very different structure, so which carries more information is not obvious.

A standard way of investigating this question is to replace different parts of speech by noise and to measure the recognition accuracy of listeners on a large set of test sentences^{2,3}.



In contrast to text, these studies show a 'vowel superiority effect'; that is, vowels seem to carry more information, because listeners perform more accurately when understanding sentences with missing consonants than with missing vowels.

The categories of vowels and consonants, however, are problematic as elemental units of information. In addition to the variation in sound structure, vowels and consonants overlap in time, so they usually can't be cleanly removed. Furthermore, the acoustic structure of one phoneme is strongly influenced by its neighbours — so much so that one phoneme often contains ample information about another^{4,5}. This has led researchers to investigate intelligibility using phoneme transitions, onsets and offsets, and other more fine-grained components⁶.

The barrier one runs up against is that traditional methods of speech analysis provide only a limited description of the complex structures in the speech signal. Stilp and Kluender¹ reasoned that one way out of this dilemma might be to estimate the information content of small speech

segments directly from the signal itself.

The information content of a code element is inversely proportional to its probability; in other words, elements that are less predictable carry more information. If the code elements were known, it would be possible to calculate information directly from the probabilities. Because they are not, Stilp and Kluender developed a simple psychoacoustic measure to estimate the potential information content of a small speech segment.

The measure, which they call cochlear-scaled spectral entropy (CSE), characterizes the degree to which one spectral slice is predictable from (or similar to) the preceding one. Similarity was defined as a simple Euclidean distance between spectral slices processed by a cochlear filter bank, which filters the sound into roughly logarithmically spaced frequency channels. The channel-centre frequencies and bandwidths are set from the psychoacoustic task of detecting pure tones in noise, and are determined largely by the spectral processing in the human cochlea.

This measure provides a step-by-step estimate of the potential information of a slice (its entropy), which the authors aggregated into either 80- or 112-millisecond segments, so that the segments could be removed in an order of CSE that was increasing, decreasing or outward from the middle, just as for letters in text. The advantage of this approach is that it makes no assumptions about speech categories and does not depend on 'hand-selected' boundaries.

If intelligibility is determined by the information content, then removing the more informative elements (and replacing them with noise) should result in a greater reduction in recognition accuracy than removing less informative elements. Results in listeners showed that CSE was a better predictor of intelligibility than the traditional categories of consonants and vowels, or consonant-vowel transitions, falling off approximately linearly with the proportion of the sentence entropy replaced.

Vowels were found to be more informative (higher CSE) on average than consonants, which can be attributed to vowels having

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higher variability in real spoken speech. This result depended crucially on the cochlear scale chosen for estimating entropy: because it is quasi-logarithmic, it gives greater weight to frequencies in the 'formant range' — that is, the locus of most vowel structure, which, contrary to what is often assumed, changes considerably over time and therefore has greater potential information. Consonants can also contain rapid change, but typically in higher frequencies, which receive less weight.

The results call into question the canonical view of speech as a sequence of consonants and vowels, and open up avenues to new statistical descriptions that might better describe its fluid nature. Approaches involving information theory have offered several insights into the signal-level sensory codes of natural images and sounds^{7,8}. These approaches work from the bottom up to identify sets of features that most compactly describe a class of signals, but are as yet insufficient to capture the highly nonlinear regularities in speech. Ultimately, getting to the bottom of speech intelligibility will require understanding not only the dimensions of variability that underlie the discriminability of speech sounds, but other informative aspects of the speech signal, such as its harmonic patterns and tonal variation, which vary from individual to individual. A further complication is that intelligibility can be adjusted dynamically to compensate for sources of noise.

Understanding the basis of speech intelligibility would have obvious benefits for speech-related technologies. Today's hearing aids already contain quite sophisticated processing, but still break down in noisy environments. Today's computers understand speech in the same way that they play chess, by brute force, and not as proficiently as humans. According to legend, a test of an early speech-recognition system goes something like this:

Computer: "Please state your four-digit personal identification number."

User: "three four five six."

Computer: "I did not understand. Please repeat."

User: "three ... four ... five ... six."

Computer: "I did not understand. Please repeat."

User: "Three ... Four ... Five ... Six."

Computer: "I did not understand. Please repeat."

User: "THREE ... FOUR ... FIVE ... SIX."

Computer: "I did not understand. Please repeat."

At which point the test user slams down the phone in frustration, and the scientists retreat back to their labs. The speaker actually did something perfectly reasonable in the face of miscommunication: spoke more clearly. Of course, when anger and frustration come into it, the message is completely lost on the computer. The irony is that what makes speech more intelligible for a human listener actually

makes matters worse for the computer.

Some day, perhaps, computers will be able to listen to us and not only get the words but also catch our drift. And some day, perhaps, we will be able to listen to our grandchildren, or even a foreign language, using hearing aids that work more like 'ear glasses' that bring clarity to a noisy world.

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GEOCHEMISTRY

Relict mantle from Earth's birth

David Graham

Geochemical evidence for the existence of the mother of all mantle-source reservoirs for volcanism has come to light. The new results have provocative implications for our understanding of Earth's interior.

Earth is a differentiated planet. Its primordial building blocks of iron-rich metal, oxides, silicate minerals, and volatile elements and compounds have been transformed over geological time into the modern-day structure of core, mantle, crust, ocean and atmosphere. Chemical and physical processes such as mantle convection, tectonic-plate recycling and magma generation through partial melting should have scrambled, if not obliterated, any coherent geochemical signature of the primordial material. Even if a vestige of such material remained, it seems unlikely that it would be found in any samples from Earth's surface or the shallow subsurface that are available to geologists.

Yet that is what new evidence suggests. On the basis of a trace-element and isotopic study, Jackson *et al.*¹ (page 853 of this issue) propose that lavas from Baffin Island, Canada (Fig. 1), were derived from a deep-Earth reservoir that has remained isolated since the earliest days of planetary accretion some 4.5 billion years ago. Their work relies on combined analyses of the isotopes of helium, lead, neodymium and hafnium. Collectively, the results are the first of their kind for terrestrial volcanic rocks.

The essence of the argument for the survival of primordial material is threefold. First, there is the occurrence of the highest ever measured ³He/⁴He ratios, in 60-million- to 62-million-year-old volcanic rocks from Baffin Island and West Greenland². The ratios are more than five times higher than values commonly observed along mid-ocean ridges that tap the shallow upper mantle. The very high ³He/⁴He signifies an elevated ratio of primordial to radiogenic noble gas — a trait that geochemists often regard as supporting the existence of a reservoir deep within Earth that is the ultimate

source for volcanic hotspots such as Hawaii and Iceland³.

Second, some of the Baffin and West Greenland lavas have lead-isotope compositions that lie on, or very close to, a 4.5-billion-year-old geochron. The geochron is a line in the ²⁰⁷Pb/²⁰⁴Pb–²⁰⁶Pb/²⁰⁴Pb diagram describing all possible parent/daughter (uranium/lead) ratios corresponding to Earth's age.

Third, the lavas have neodymium isotope compositions (¹⁴³Nd/¹⁴⁴Nd) consistent with derivation from a mantle source comprised of primordial silicates. On the basis of the landmark discovery of positive terrestrial ¹⁴²Nd anomalies⁴, this mantle source has a samarium/neodymium ratio that is about 5% larger than the ratio in chondrites. (Chondrites are stony meteorites that formed during the earliest stages of Solar System development and were never melted after their formation. They are thought to closely resemble the building blocks involved in Earth's formation.) Because ¹⁴²Nd was produced by the extinct radioactive decay of ¹⁴⁶Sm (half-life of 103 million years), the excess of ¹⁴²Nd relative to chondrites in all terrestrial samples seems to require the formation, early in Earth's history, of an 'early depleted reservoir' with a high Sm/Nd ratio as a residue of partial melting. Negative ¹⁴²Nd anomalies have not been found in terrestrial rocks, so this early depleted reservoir represents the ultimate ancestor of all other mantle-source reservoirs for volcanism.

These three fundamentally coherent isotopic characteristics of helium, lead and neodymium have now been found together for the first time. Notably, they occur in contemporaneous primitive lavas from Baffin Island and West Greenland that erupted during the opening of the North Atlantic. The results are evidence that a